MIL-B-60073(MU)
18 June 1964
(See section 6)

MILITARY SPECIFICATION

BOMB, NONPERSISTENT GAS, SPHERICAL, 4.5 INCH, M139; FILLING AND ASSEMBLY

1. SCOPE

1.1 This specification covers the filling and assembling of one type of spherical bomb.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWING

EDGEWOOD ARSENAL

DL14-5-2768 - Bomb, Nonpersistent Gas, Spherical, 4.5 Inch, M139 - Filling and Assembly.
(All drawings referenced in this specification are listed on above drawing.)

FSC 1325

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain prior to their use that all materials and components have met their respective specifications.

3.2 Filling. The bomb shell halves shall be filled with the chemical agent specified on Drawing C14-5-2768 and shall be within the limits shown in table I when determined as specified in 4.6.3 (see 6.3 and 6.4). The distance "d" is defined on Drawing C14-5-2768.

Table I. Filling level

Temperature range °F	"d" height in inches	
	Minimum	Maximum
Between 40 and 50	0.56	0.65
Between 50 and 60	0.54	0.63
Between 60 and 70	0.53	0.62
Between 70 and 80	0.51	0.60
Between 80 and 90	0.50	0.59

3.2.1 Filling level. When an alternative procedure is approved for use to determine the filling level, the procedure shall be monitored in accordance with 4.6.3 to insure the requirement of this specification is satisfied.

3.3 Closing. The bomb shell halves shall be closed as shown on Drawing C14-5-2768. From the time of filling until completion of the test in accordance with 4.6.2, the shell halves shall be kept in a closure-up position (see 6.4).

3.4 Agent contamination. The exposed surfaces of the bomb shall be free from GB contamination (see 4.6.1 and 6.5).

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3.5 Helium leakage. The bomb shell halves shall not leak helium through the closure at a rate greater than 1×10^{-6} atmospheric cubic centimeter (cc) per second when tested in accordance with 4.6.2.

3.6 Assembly. The bomb shall be assembled as shown on Drawing C14-5-2768.

3.7 Preproduction. When specified in the contract or order, a preproduction sample of bombs shall be produced in accordance with this specification for examination and test.

3.8 Workmanship. The bombs shall be free from foreign matter and visible damage such as dents and cracks which could affect their intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Supplier's responsibility will be as specified in the contract or order.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedures which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 Substation inspection. If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of Standard MIL-STD-105 are applicable to each substation and to the end item.

4.3 Lotting. A lot shall consist of the bombs filled by one manufacturer, on one filling line under essentially the same filling conditions. However, no more than two lots of boosters, bursters, agent or fuzes shall be represented in any one lot of filled and assembled bombs.

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4.4 Sampling. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.5 Inspection provisions.

4.5.1 Examination. Sample bombs shall be examined in accordance with the classification of defects and with Standard MIL-STD-105.

4.5.2 For critical defects. Each bomb in the lot shall be inspected for its critical characteristics listed in the classification of defects.

4.5.3 Classification of defects.

4.5.3.1 Bomb, nonpersistent gas, spherical, 4.5 inch, M139 - assembly (Dwg C14-5-2768).

Categories	Defects	Acceptance standards
Critical:		
1	Contamination	4.6.1
2	Helium leakage	4.6.2
3	Filling excessive	4.6.3
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Shell halves loose after crimping fastener	
103	Booster, burster or fuze loose after assembly	
104	Marking illegible or incorrect	
105	Workmanship (3.8)	
Minor:	AQL 2.5 percent defective	
201	Filling insufficient	

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4.6 Tests.

4.6.1 GB contamination. Contamination shall be determined by the use of an approved Edgewood Arsenal GB agent detector system.

4.6.2 Helium leakage. Prior to closing, the void above the agent shall be evacuated to a pressure not greater than 2 pounds per square inch absolute (psia) and filled with helium to an atmospheric pressure of 15 psia to insure a partial pressure due to helium of not less than 10 psia after closing. The closure area shall be sealed off by a test head manifolded to a vacuum system and a mass spectrometer type helium leak detector. The test head shall be evacuated to and maintained at a pressure lower than 20 microns of mercury during the test. The leak detector shall be calibrated to respond to a leak of 1×10^{-6} atmospheric cc per second of helium at the test head when a standard leak is introduced into the system at that point.

4.6.3 Filling level. The height of the GB in the filled shell half shall be located radiographically in accordance with Edgewood Arsenal Inspection Standard R1-2 with the shell half in the identical position in which it was filled and closed (see 6.2).

5. PREPARATION FOR DELIVERY

Packaging, packing and marking information is not applicable to this specification.

6. NOTES

6.1 Intended use. This item is used as the agent container for chemical filled rockets.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The Government will furnish the radiographic inspection standard.
(c) Preproduction when required.

1. Quantity required.
2. Time allowed for suppliers submission of samples for Government test and evaluation after award of contract.
3. Destination of test facility when testing is performed by the Government.
4. Packing and shipping instructions when applicable.
5. Time required for the Government to notify the supplier whether or not to proceed with production.
6. Conditions when preproduction will not be required.

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6.3 Filling level. When filled as specified in 3.2, the shell half carries approximately 0.6 pounds of chemical agent. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Precautions. Because of the hazards attending the filling of munitions with GB and assembling of explosives, suppliers are advised to exercise all necessary precautions to protect personnel from injury during the filling and handling of munitions. Care should also be taken to prevent wetting the closure with agent thereby making void the results of the helium leak test.

6.5 Decontamination. Decontamination can be accomplished by subjecting the bomb to a spray of decontaminating solution (5-to 10-percent commercial grade soda ash, Na_2CO_3) for a minimum period of 1 minute at a minimum pressure of 15 psig in such a manner as to completely wash all surfaces. The bomb should then be rinsed with water to remove all traces of the decontaminating solution. The temperature of the spray and rinse water should be maintained at a range of 140° to 160°F. Spray pressures should have gages with an accuracy of 1 psig. An approved procedure of control of the decontaminating solution strength should be maintained by periodic examination and the time the bomb is subjected to the spray should be automatically controlled and recorded.

6.6 Safety classification. This item carries the following safety classification: Chemical group A (special) explosive hazard class 4, compatibility class A.

6.7 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description 196-131-749, dated 15 March 1962.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1325-A128

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